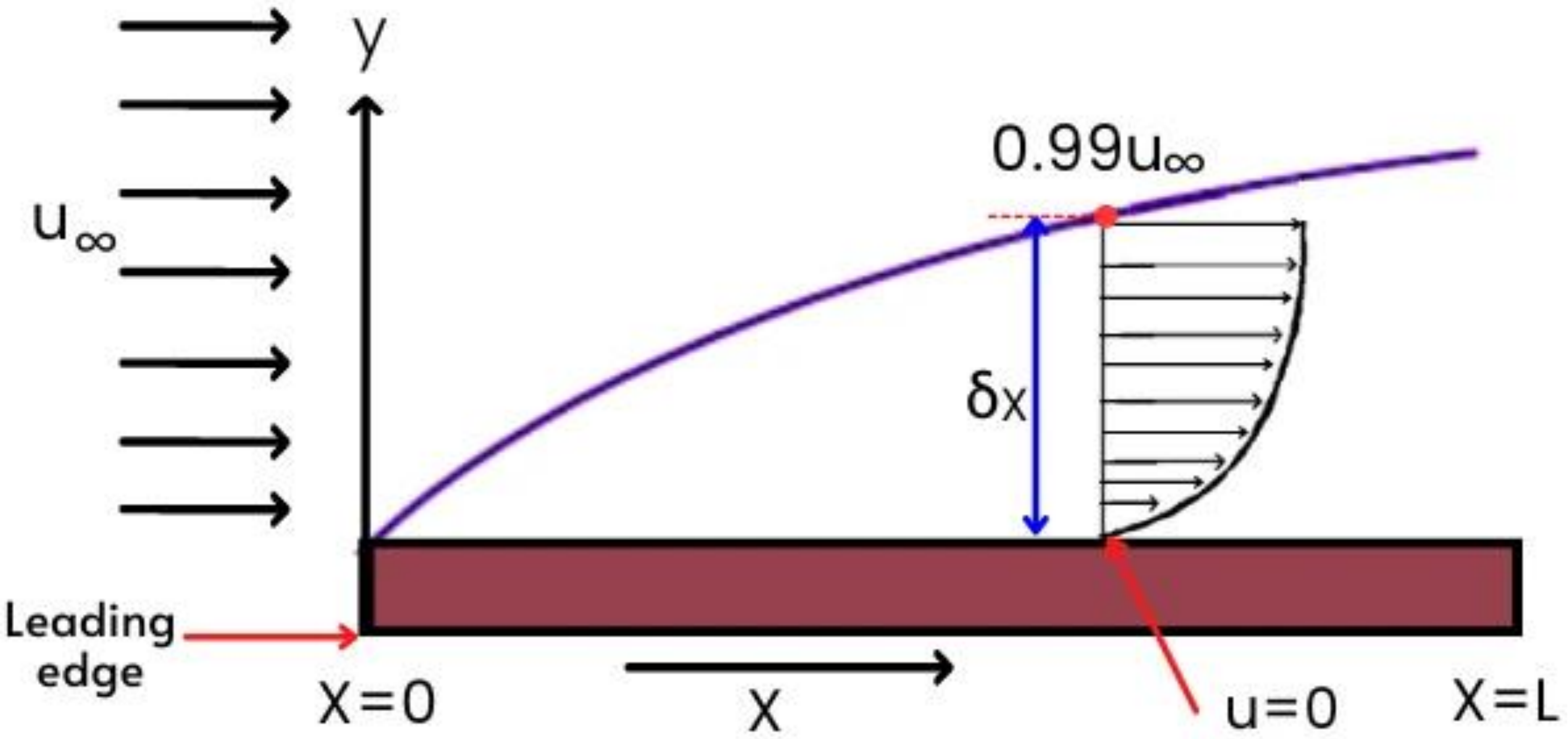
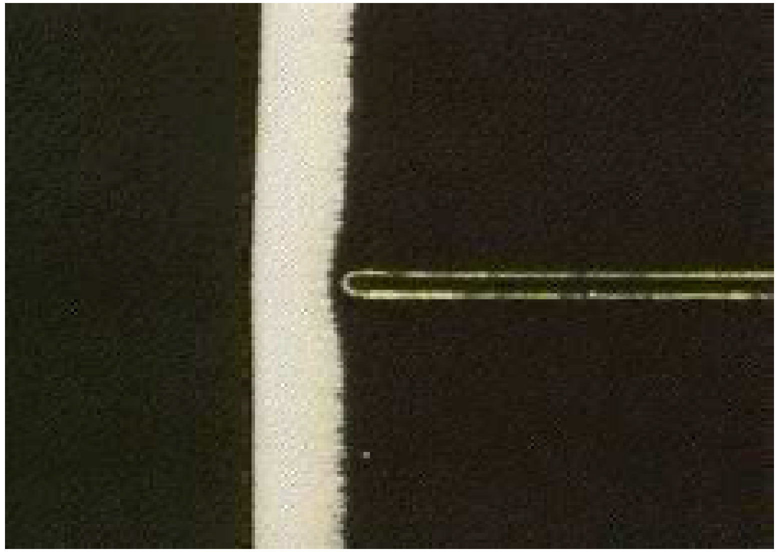
BOUNDARY LAYER

- Whenever there is relative movement between a fluid and a solid surface, whether externally round a body, or internally in an enclosed passage, a boundary layer exists with viscous forces present in the layer of fluid close to the surface.
- Boundary layers can be either laminar or turbulent.
- A reasonable assessment of whether the boundary layer will be laminar or turbulent can be made by calculating the Reynolds number of the local flow conditions.
- Flow separation or boundary layer separation is the detachment of a boundary layer from a surface into a wake.
- Separation occurs in flow that is slowing down, with pressure increasing, after passing the thickest part of a streamline body or passing through a widening passage

Thickness of Boundary Layer



Viscous boundary layer



- An originally laminar flow is affected by the presence of the walls.
- Flow over flat plate is visualized by introducing bubbles that follow the local fluid velocity.
- Most of the flow is unaffected by the presence of the plate.
- However, in the region closest to the wall, the velocity decreases to zero.
- The flow away from the walls can be treated as inviscid, and can sometimes be approximated as potential flow.
- The region near the wall where the viscous forces are of the same order as the inertial forces is termed the boundary layer.
- The distance over which the viscous forces have an effect is termed the boundary layer thickness.
- The thickness is a function of the ratio between the inertial forces and the viscous forces, i.e. the Reynolds number. As Re increases, the thickness decreases.

Thickness of Boundary Layer

Displacement Thickness

$$V\delta^* = \int_0^\infty (V - v)\, dy$$

$$\delta^* = \int_0^\infty \left(1 - \frac{v}{V}\right) dy$$

Momentum Thickness

$$\rho V^2 \theta = \rho \int_0^\infty (V - v)\, v\, dy$$

$$\theta = \int_0^\infty \frac{v}{V}\left(1 - \frac{v}{V}\right) dy$$

Energy Thickness

$$\frac{1}{2}\rho V^3 \delta_E = \frac{1}{2}\rho \int_0^\infty (V^2 - v^2) v\, dy$$

$$\delta_E = \int_0^\infty \frac{v}{V}\left(1 - \frac{v^2}{V^2}\right) dy$$

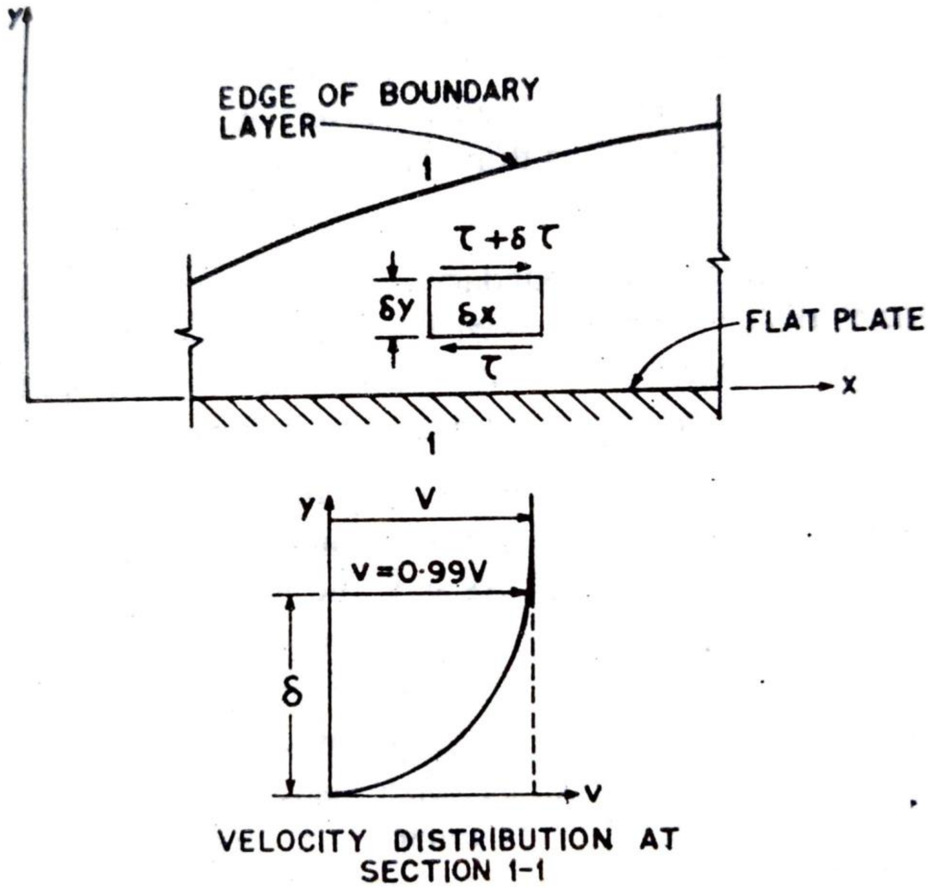
Boundary Layer Equation

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial x}$$

$$u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial y}$$

Boundary Layer Equation



$$-\tau\delta x$$

$$(\tau + \partial \tau)\,\delta x = \left[\tau + \frac{\partial \tau}{\partial y}\delta y\right]\delta x$$

Boundary Layer Equation

$$\frac{1}{\rho}\frac{\partial \tau}{\partial y}$$

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \frac{1}{\rho}\frac{\partial \tau}{\partial y}$$

Boundary Layer Equation

$$\tau = \mu \frac{\partial u}{\partial y}$$

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \upsilon \frac{\partial^2 u}{\partial y^2}$$

Boundary Layer Equation

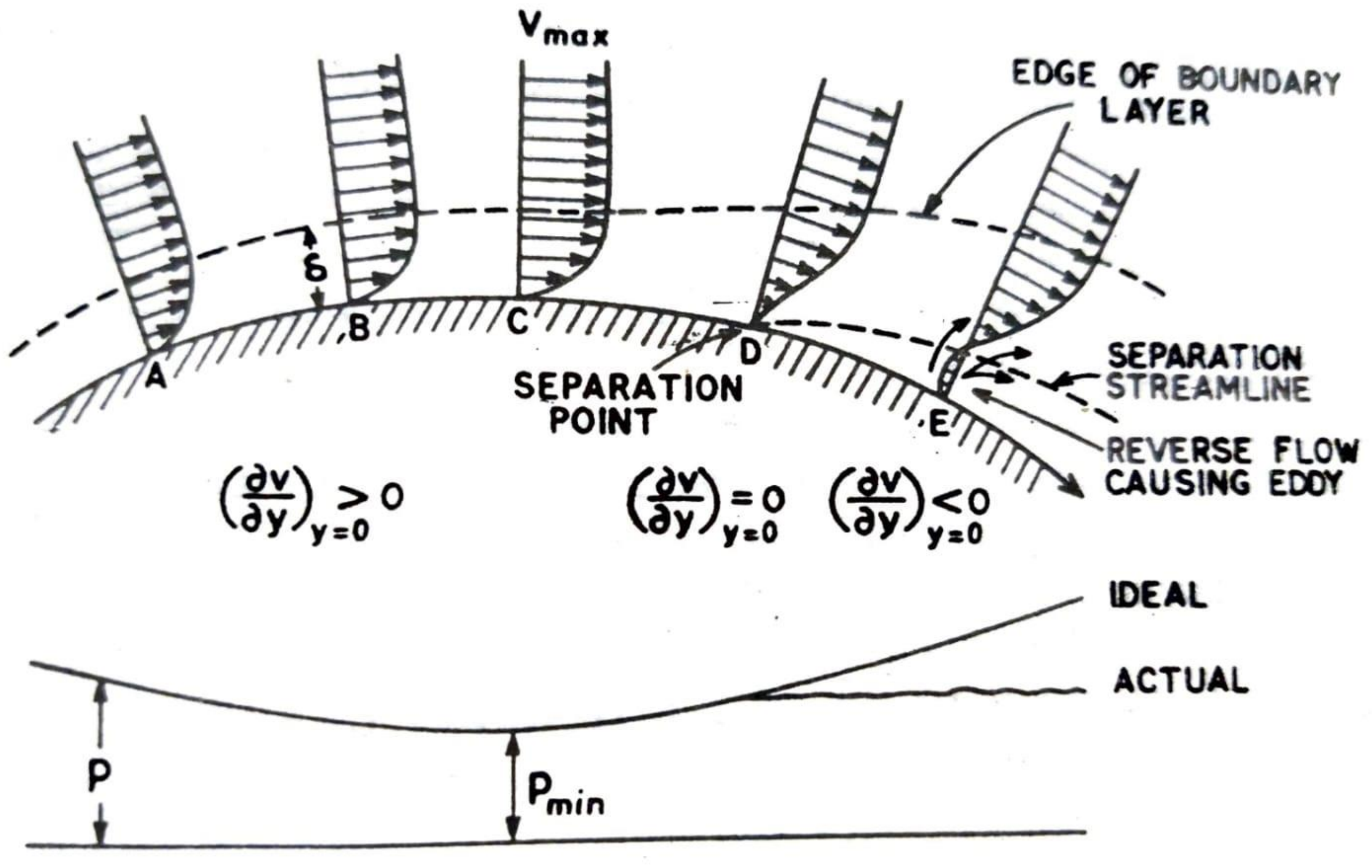
$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \upsilon\frac{\partial^2 u}{\partial y^2}$$

$$-\frac{1}{\rho}\frac{\partial p}{\partial y} = 0$$

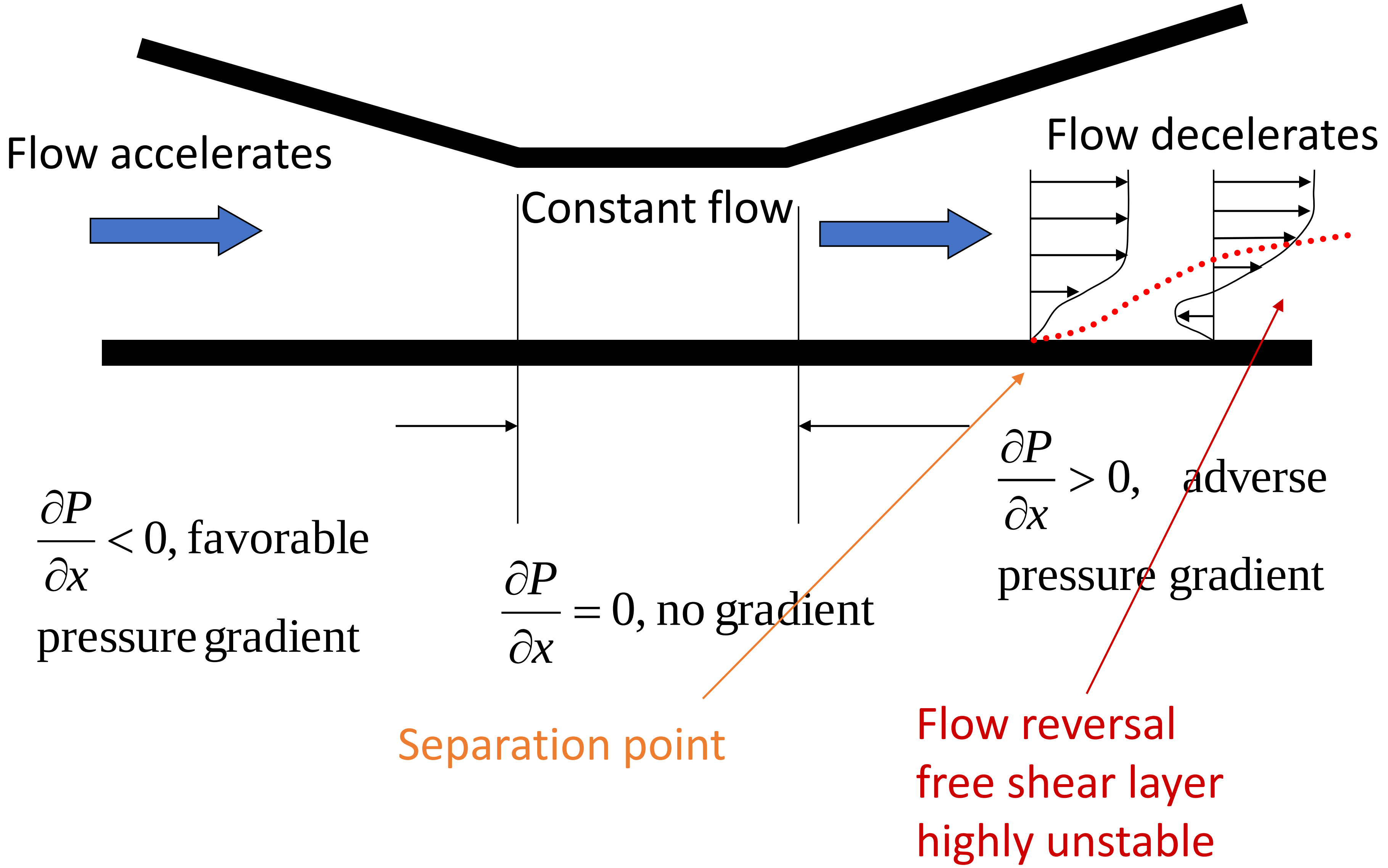
Prandtl’s boundary layer equation

Separation of Boundary Layer



PRESSURE DISTRIBUTION ALONG THE BOUNDARY

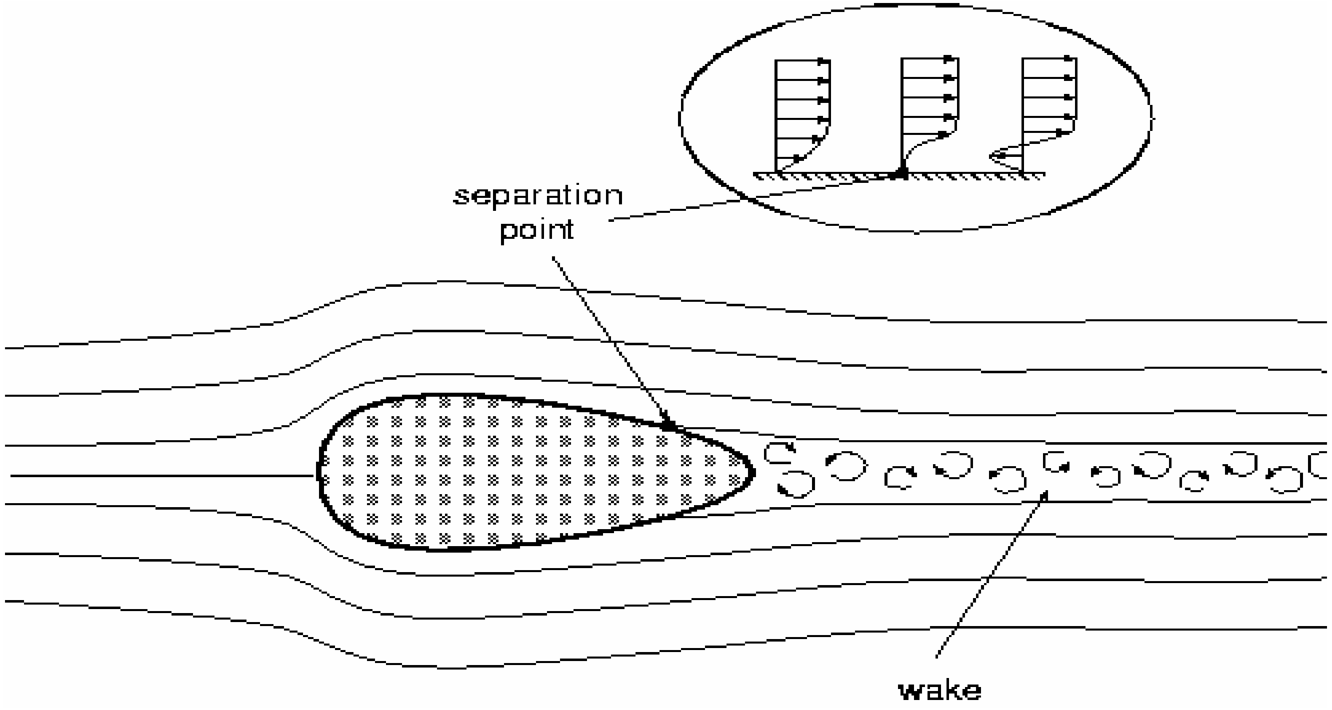
Boundary Layer and separation



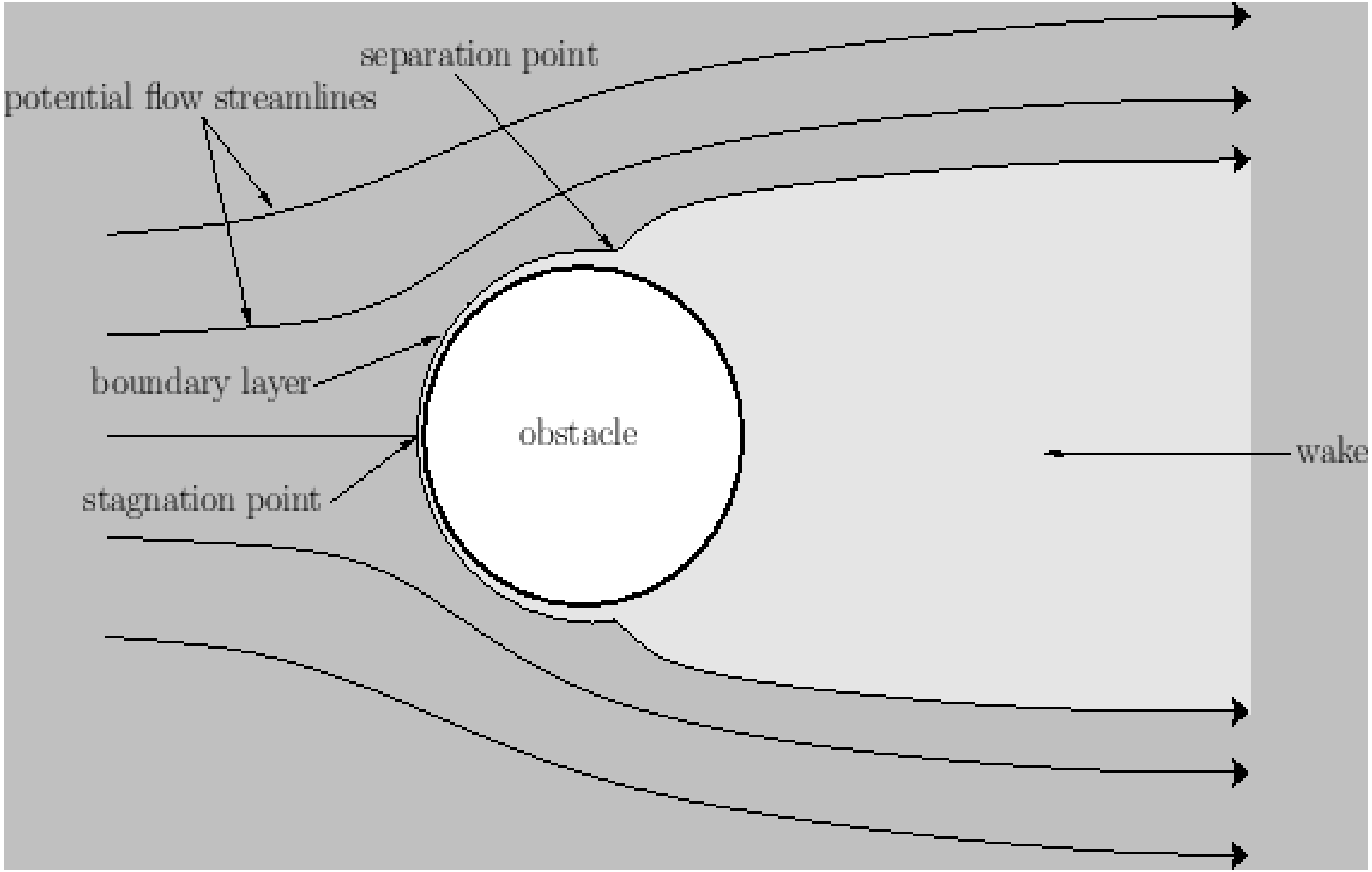
Flow separation

Flow separation occurs when:

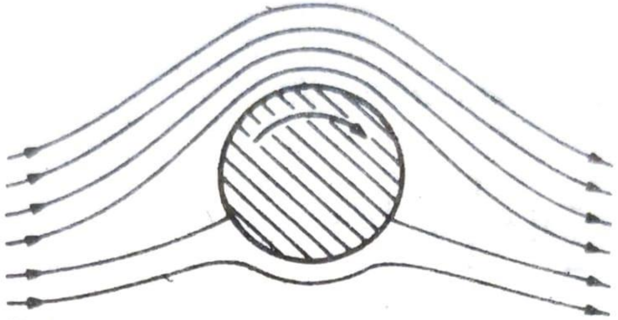
- ❖ The velocity at the wall is zero or negative and an inflection point exists in the velocity profile,
- ❖ A positive or adverse pressure gradient occurs in the direction of flow.



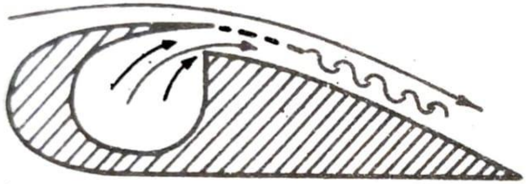
- Flowing against an increasing pressure is known as flowing in an adverse pressure gradient.
- The boundary layer separates when it has travelled far enough in an adverse pressure gradient that the speed of the boundary layer relative to the surface has stopped and reversed direction.
- The flow becomes detached from the surface, and instead takes the forms of eddies and vortices.
- The fluid exerts a constant pressure on the surface once it has separated instead of a continually increasing pressure if still attached
- In aerodynamics, flow separation results in reduced lift and increased pressure drag, caused by the pressure differential between the front and rear surfaces of the object.


separation point
potential flow streamlines
boundary layer
obstacle
wake
stagnation point

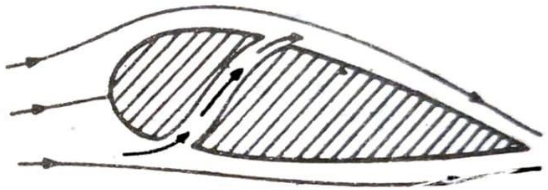
Methods of controlling the Boundary Layer



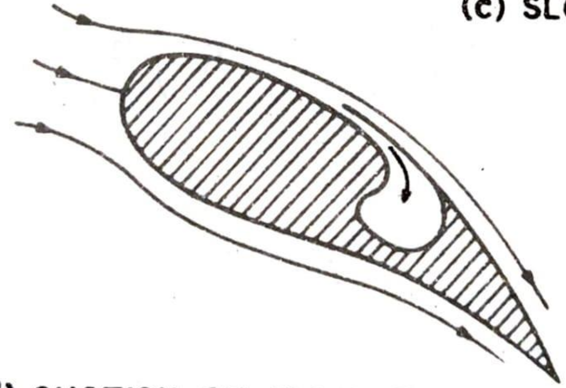
(a) FLOW PAST A ROTATING CYLINDER



(b) INJECTING FLUID INTO BOUNDARY LAYER



(c) SLOTTED WING



(d) SUCTION OF FLUID FROM BOUNDARY LAYER

THANK YOU